OBJECT DETECTION MODEL TO BE TRAINED AND COMBINED WITH DEPTH ANALYSER

ABSTRACT

This project involves the development of an object detection model integrated with a depth analyzer to enhance spatial understanding in 3D environments. By combining object detection techniques with depth information, the system aims to improve the accuracy of object localization and scene interpretation

TECHNOLOGIES USED

- **Machine Learning Model:** YOLOv7 for object detection optimized for real-time performance.
- **Depth Analysis:** Depth analyzer for improving spatial understanding in 3D environments
- **Framework:** TensorFlow Lite for lightweight ML model deployment and OpenCV for comprehensive computer vision tasks on edge devices.

APPLICATIONS USED

- **Autonomous Vehicles:** Enhancing navigation and collision avoidance for self-driving cars.
- **Robotics:** Enabling real-time obstacle detection and navigation in complex environments
- **Assistive Technologies:** Helping visually impaired individuals by detecting obstacles in their path.